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Prescribing to tumor apex in episcleral plaque iodine-125 brachytherapy for medium-sized choroidal melanoma: A single-institutional retrospective review

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ABSTRACT

PURPOSE: To report an institutional experience with episcleral plaque brachytherapy for medium-sized uveal melanoma. Variations in prescription dose point and dose rate were compared with Collaborative Ocular Melanoma Study (COMS) Group.

METHODS AND MATERIALS: A retrospective review was performed for 116 patients treated with iodine-125 plaque brachytherapy. About 85 Gy was prescribed to either the tumor apex (108 patients) or at 5 mm (8 patients) with dose rate ranging from 50.6 to 98.2 cGy/h. Patients were followed up for local tumor control, eye preservation, and vision retention. Dose and dose rate to tumor and sensitive structures were calculated. Multivariate and univariate analyses were performed to investigate correlation between clinical outcomes and dose/dose rate variables.

RESULTS: Patients in this study were slightly older with worse visual acuity at baseline, but tumor size and position and ratio of ciliary body involvement were comparable to COMS population. Outcomes data were comparable to COMS: 95.3% local tumor control at 5 years and 77.7% vision preservation at 3 years. Only 4 patients needed enucleation because of tumor growth. Significant correlation was found between enucleation and tumor height and maximal scleral dose/dose rate as well as vision retention and tumor height and macula dose/dose rate.

CONCLUSIONS: For tumors with <5 mm height, prescribing to tumor apex enabled to decrease dose to all sensitive structures without any loss of local control. Although dose rate was lowered to 50.6 cGy/h from the American Brachytherapy Society guidelines (60–105 cGy/h) because of limited availability of operating room (i.e., weekly), there was no difference in either local tumor control or complications.

Keywords: Uveal melanoma; Episcleral plaque; ^{125}I; Dose rate

Introduction

Uveal melanoma, although rare, is the most common primary intraocular malignancy in adults and represents a threat to both vision and life of the patient. Treatment options for uveal melanoma include enucleation, photocoagulation, eyewall resection, external beam radiotherapy (protons), and plaque brachytherapy (1–5). Historically, large tumors (>10 mm in height or >16 mm in the largest basal diameter) have been treated with enucleation. Gamma Knife radiosurgery has been also considered as an alternative treatment option to enucleation for large uveal melanoma (6). Small tumors (≤2.5 mm in height and ≤10 mm in the largest dimension) may be followed closely with most patients demonstrating no tumor growth (7).

For progressive small tumors and medium-sized tumors, some form of treatment is recommended. Enucleation provides excellent control at the cost of poor cosmesis and

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